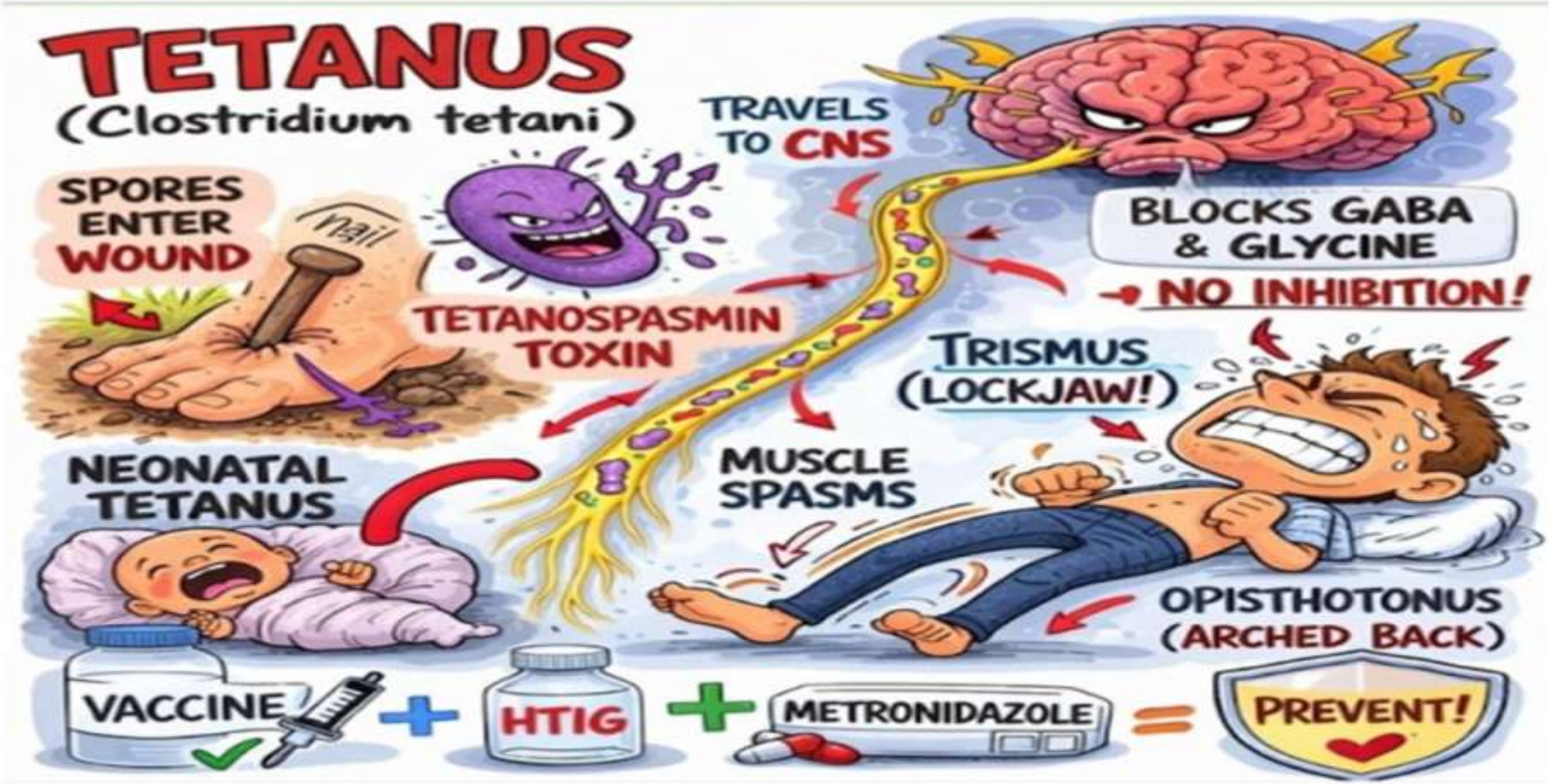


By

ASS. Prof. Dr. Tasneem Mohammed Ali

Learning Objectives

- Understand global epidemiology of tetanus
- Describe transmission and pathogenesis
- Identify clinical features
- Review prevention and PEP
- Understand neonatal tetanus prevention

Introduction

Acute, potentially fatal disease caused by Clostridium tetani

Vaccine-preventable but persists in developing countries

Not transmitted person-to-person

Global Epidemiology

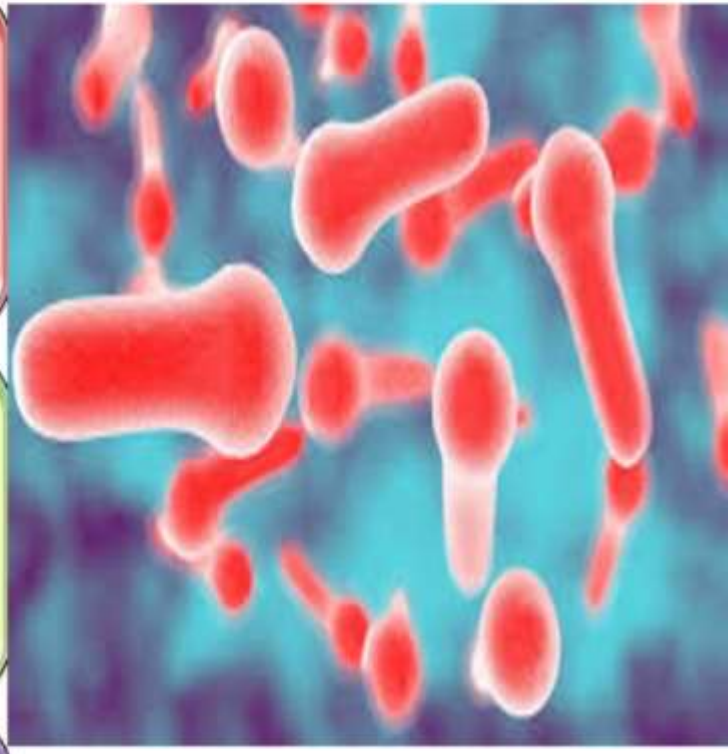
- Occurs worldwide; more common in hot, humid regions
- Decline in developed nations due to vaccination
- cases globally 11,367~ :2014
- reported cases9 :2013Egypt

Etiology

Clostridium tetani: Gram-positive, anaerobic, spore-forming bacillus

Terminal spores give drumstick appearance

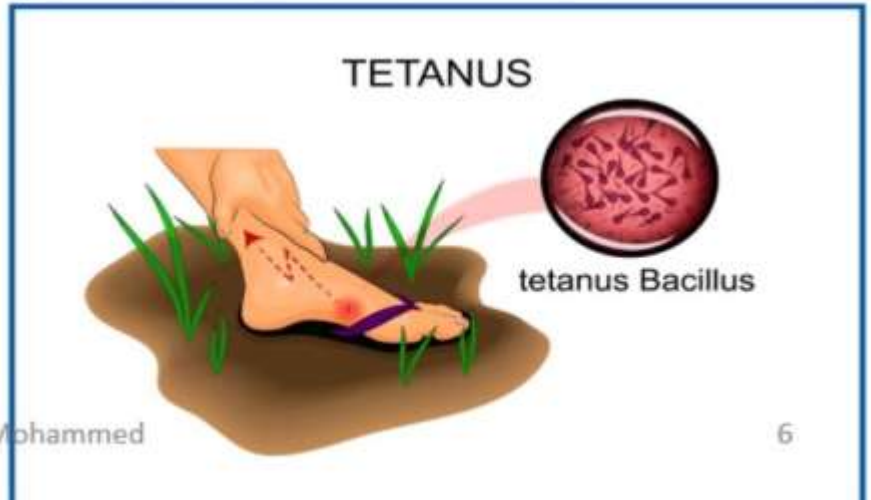
Highly resistant to heat and antiseptics



Reservoir

Soil, intestines of animals and humans

Spores survive long in environment



Tetanus

Mode of Transmission

Wound Contamination with Spores of *Clostridium tetani*

Common Sources:

- ✓ Puncture Wounds
- ✓ Burns
- ✓ Crush Injuries
- ✓ Childbirth

✓ Crush Injuries

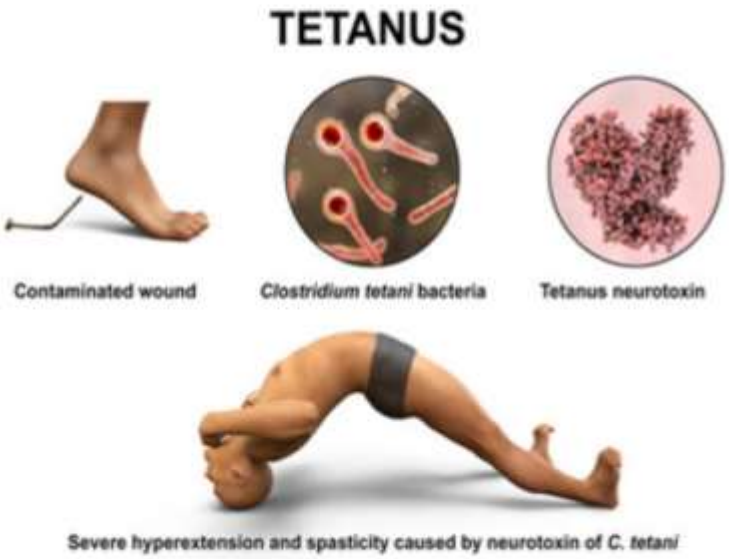
✓ Burns

✓ Childbirth

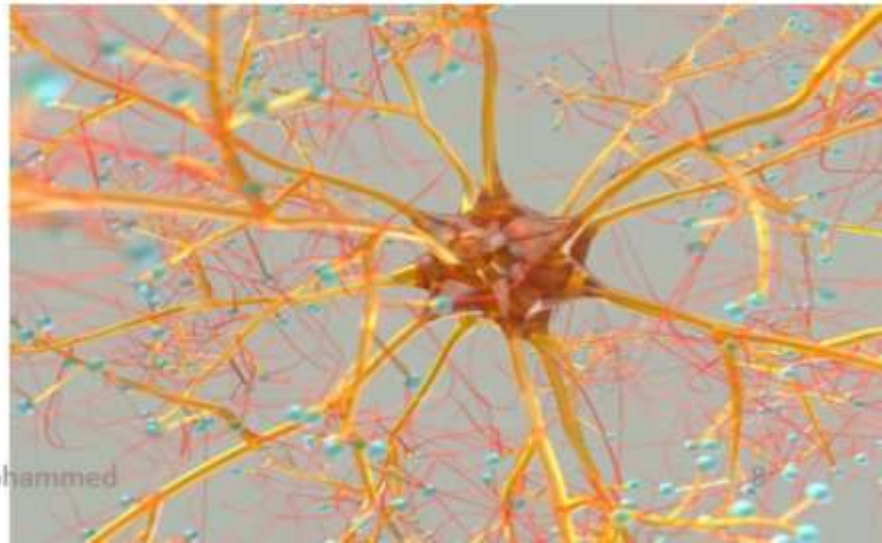
Spores of *Clostridium tetani* infect wounds, commonly found in soil, dust, and animal feces.

Pathogenesis

Tetanospasmin neurotoxin blocks inhibitory neurotransmitters



Leads to muscle rigidity and spasms

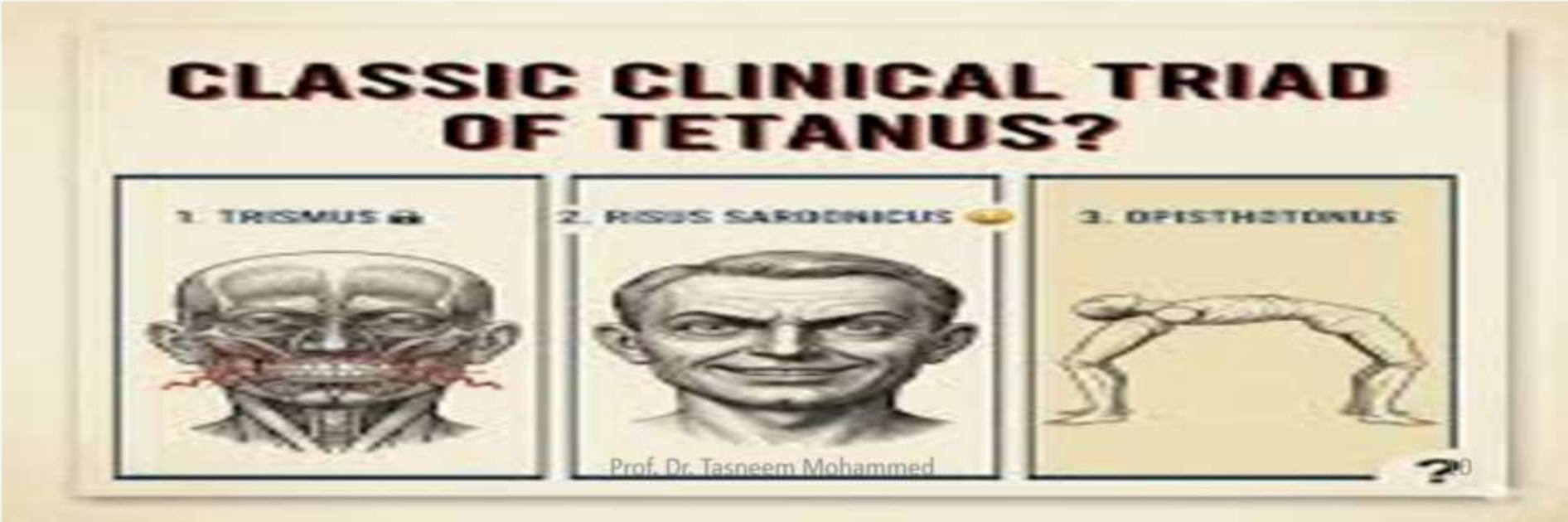


Clinical Forms

Generalized Tetanus

Trismus, risus sardonicus, neck rigidity

Opisthotonos, autonomic instability



Generalized tetanus:

- It is the most common type (about 80%) of reported tetanus. The disease usually presents with a descending pattern.

- Early signs and symptoms include stiffness or cramps in muscles around the wound.

- The disease usually presents with a descending pattern where face is first affected with facial pain, change in facial expression (risus sardonicus), stiffness of jaw (trismus) or lockjaw, followed by stiffness of the neck, difficulty in swallowing, and rigidity of abdominal muscles.

reanimatio

Risus sardonicus: Sustained contraction of facial musculature produces a sneering grin expression known as risus sardonicus.

Risus sardonicus (spasm of frontalis & muscles of angles of mouth)

LOS ANG

Cephalic tetanus:

It is a rare form of the disease

occasionally occurring with otitis media (ear infections) in which C. tetani is present in the flora of the middle ear,

or following injuries to the head.

There is involvement of the cranial nerves, especially in the facial area.

Cephalic tetanus:

Spasm of larynx, diaphragm, and intercostal muscles may result in acute respiratory failure.

Sudden contraction of muscle groups may result in opisthotonos.

Instability of autonomic nervous system is common with a temperature rise of 2°- 4°C above normal, sweating,

elevated blood pressure, and episodic rapid heart rate.

Spasms may occur frequently and last for several minutes. Spasms continue for 3-4 weeks.

Complete recovery may take months.

Neonatal tetanus

form of generalized tetanus that occurs in newborn infants.

Neonatal tetanus occurs in infants born **without protective passive immunity,** because the mother is not immune.

It usually occurs through **infection of the unhealed umbilical stump,** particularly when the stump is cut with an unsterile instrument. Neonatal tetanus is common in some developing countries

Complications

- Respiratory failure
- Fractures
- aspiration pneumonia
- sepsis

Diagnosis

Clinical diagnosis

lab confirmation not required

PERIOD OF COMMUNICABILITY:

- **Tetanus is not contagious from person to person. It is the only vaccine-preventable disease that is infectious.**
- **Spores may remain viable for many years in the environment.**

SUSCEPTIBILITY AND RESISTANCE:

- **Susceptibility is general.**
- **Active immunity is induced by tetanus toxoid and persists for at least 10 years after full immunization.**
- **Transient passive immunity follows injection of tetanus immune globulin (TIG) or tetanus antitoxin .**
- **Infants of actively immunized mothers acquire passive immunity that protect them from neonatal tetanus**
- **Recovery from tetanus may *NOT* result in immunity or recovery from tetanus should be followed by active immunization as second attack can occur.**

✓PREVENTION AND CONTROL

PRIMARY PREVENTION

- **Educate the public on:**

1. **the necessity for complete immunization with tetanus toxoid,**
2. **the hazards of puncture wounds and closed injuries that are particularly liable to be complicated by tetanus, and the potential need after injury for active and/or passive prophylaxis.**

- **Community development and clean environment.**
- **Replacing animal carts by mechanical vehicles.**
- **Mechanization of agricultural activities in rural areas.**

Specific prevention

- **Passive immunization by :**
- **Tetanus antitoxin, anti-tetanic serum, equine anti-toxine.**
- **Human antitoxic immunoglobulins (HTIG).**
- **They are used for temporary protection against infection.**

Active immunization:

Tetanus toxoid adsorbed is available combined with diphtheria toxoid and acellular pertussis (DTaP), or combined with diphtheria toxoid and whole cell pertussis vaccine (DTP).

Adsorbed preparations without pertussis antigen are available as (DT) for use in children less than 7 years or (Td) for use in adults.

Tetanus vaccine also is available as monovalent tetanus toxoid (TT).

Tetanus toxoid should be given for:

- Infants and young children.
- Pregnant women.
- **High risk groups:**
- Workers in contact with soil, sewage and domestic animals.
- Members of military forces and police men.
- Older adults who are currently at higher risk for tetanus and tetanus mortality.
- Post exposure prophylaxis (injury in suspected tetanus spores contaminated places).
- All travelers should be up to date with the vaccine before departure.
- Adolescents and adults who are not actively immunized before.

Tetanus Vaccination Schedule (WHO-Aligned)

Population Group	Recommended Doses	Schedule / Timing	Notes
(WHO childhood schedule)	**5 doses total**	**Primary series:** 3 doses of DPT at **2, 4, 6 months** **Boosters:** at **18 months**, **4–7 yrs**, and **12–15y**	Ensures long-term protection across childhood and adolescence.
Unimmunized individuals ≥7 years	**3-dose primary series**	Given as **three spaced doses** (e.g., 0, 1 month, 6–12 months)	Used when no previous tetanus immunization exists.
Severely immunocompromised individuals (including HIV)	**Same as immunocompetent persons**	Follow the **same age-appropriate schedule** (childhood 5-dose program or 3-dose adult primary series)	Immune response may be **suboptimal**, but vaccination is still indicated.

Post exposure immunization (Prophylaxis in wound management):

proper wound management: wound cleaning and, where required, surgical debridement and the proper use of antibiotics.

Tetanus prophylaxis in patients with wounds is based on careful assessment of whether the wound is clean or contaminated and the immunization status of the patient.

Post-Exposure Immunization

Prophylaxis in Wound Management

Fully Immunized

Minor, Clean Wound

Booster dose of Tetanus Toxoid if > 10 years since last dose

Major or Contaminated Wound

Booster dose of Td (if > 5 years since last dose)

Not Fully Immunized

Minor, Clean Wound

Tetanus Toxoid Dose

Major or Contaminated Wound

Tetanus Toxoid Dose

Tetanus Toxoid + Human TIG + + Booster Dose

For Major or Contaminated Wound

Tetanus Immunoglobulin + Tetanus Toxoid

MANAGEMENT STRATEGIES AND CLINICAL DECISION FLOW

- **Airway Protection**
- Airway management is critical due to spasms risking ventilation; intubation or tracheostomy may be required.
- **Toxin Neutralization**
- Administration of human tetanus immune globulin binds unbound toxin and improves patient prognosis.
- **Infection Control**
- Antibiotic therapy and wound debridement reduce bacterial growth and toxin production effectively.
- **Symptom Stabilization and Support**
- Muscle spasm management, autonomic control, and ICU supportive care stabilize the patient

Tetanus Management

Airway Protection

Airway management is critical due to spasms risking ventilation; intubation or trachestomy may be required.

Toxin Neutralization

Administration of human tetanus immune globulin binds unbound toxin and improves patient prognosis.

Infection Control

Antibiotic therapy and wound debridement reduce bacterial growth and toxin production effectively.

Symptom Stabilization and Support

Muscle spasm management, autonomic control, and ICU supportive care stabilize the patient.

Secondary prevention

Control of patient:

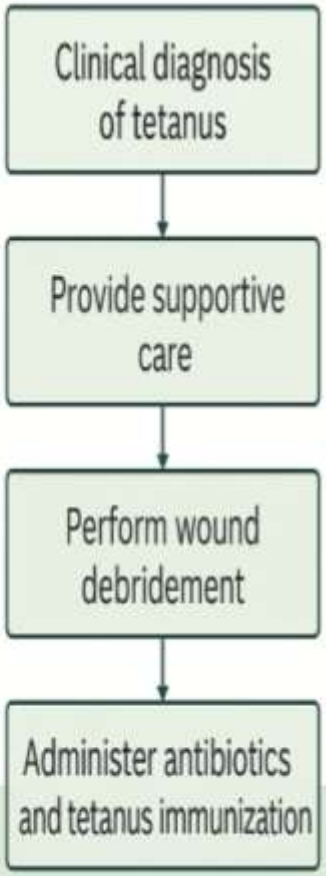
1) **Report to local health authority.**
2) **Specific treatment:**

- **TIG IM in doses of 3,000-6,000 IU.**
- **If TIG is not available, tetanus antitoxin (equine origin) in a single large dose should he given IV following appropriate testing for hypersensitivity;**
- **IV metronidazole in large doses .should lie given for 7-14 days.**

Management

Management

- Supportive care
- Wound debridement
- Antibiotics (e.g., metronidazole)
- Tetanus immunization



- **The wound should be described widely and excised if possible. Wide debridement of the umbilical stump in neonates is not indicated.**

- **Maintain an adequate airway and employ sedation as indicated; muscle relaxant claims together with tracheostomy or nasotracheal intubation and mechanically assisted respiration may be life saving.**

- **Active immunization should be initiated concurrently with therapy.**

3) Investigation of contacts and source of infection: Case investigation to determine circumstances of injury.

EPIDEMIC MEASURES:

- In the rare outbreak, search for contaminated street drugs.

Disaster implications:

- Social upheaval (military conflicts, riots) and natural disasters (floods, hurricanes, earthquakes) that cause many traumatic injuries in non immunized populations will result in an increased need for TIG or tetanus antitoxin and toxoid for injured patients.

INTERNATIONAL MEASURES:

- Up-to-date immunization against tetanus is advised for international travelers.

TETANUS NEONATORUM

- It is a serious health problem in many developing countries where maternity care services are limited and immunization against tetanus is inadequate.

- Most newborn infants with tetanus have been born to non immunized mothers delivered by a traditional birth attendant outside a hospital.

- Tetanus neonatorum declined considerably in many developing countries due to immunization of women in child bearing period with tetanus toxoid.

- WHO estimates that in 2013 , 49,000 newborns died from NT, a 94% reduction from the situation in the late 1980s.

- **The disease usually occurs through introduction via the umbilical cord of tetanus spores during delivery through the use of an unclean instrument to cut the cord, or after delivery by "dressing" the umbilical stump with substances heavily contaminated with tetanus spores, frequently as part of natal rituals.**

- **In neonates, inability to nurse is the most common presenting sign.**

- **Tetanus neonatorum is typified by a newborn infant who sucks and cries well for the first few days after birth and subsequently develops progressive difficulty and then inability to feed because of trismus, generalized stiffness with spasms or convulsions and opisthotonus.**

- **The average incubation period is about 6 days, with a range from 3 to 28 days.**

- **Overall, neonatal tetanus case-fatality rates are very high; exceeding 80% among cases with short incubation periods.**

Neonatal Tetanus Prevention

Immunization of women of child-bearing age

Skilled attendance at delivery

Aseptic umbilical cord care

Five doses protect throughout reproductive years

Prevention of tetanus neonatorum

can be achieved by a combination of two approaches:

1) **by improving maternity care with emphasis on increasing the tetanus toxoid immunization coverage of women of childbearing age (especially pregnant women), and**
2) **by increasing the proportion of deliveries attended by trained attendants.**

- **Important control measures include:**
- **Licensing of midwives;**
- **Providing professional supervision and education as to methods, equipment and techniques of asepsis in childbirth;**

- **Educating mothers, relatives and attendants in the practice of strict asepsis of the umbilical stump of newborn infants.**

- **This is especially important in many lesser developed areas where strips of bamboo are used to the umbilical cord and where ashes, cow dung or other contaminated substances are traditionally applied to the umbilicus.**

Tetanus Toxoid (TT) Schedule for Pregnant Women

Dose	Time of Administration	Protection Provided
TT1	At first contact or as early as possible during pregnancy	No reliable protection yet
TT2	At least **4 weeks after TT1**	Protects mother and newborn for **1–3 years**
TT3	At least **6 months after TT2**	Protection for **at least 5 years**
TT4	At least **1 year after TT3**	Protection for **at least 10 years**
TT5	At least **1 year after TT4**	Protection for **childbearing age (lifetime reproductive period)**

Tetanus Toxoid Vaccination for Pregnant Women

Woman's Vaccination Status	Doses Needed	Timing of Doses
No previous tetanus vaccination	TT1 + TT2 (minimum)	TT1 during pregnancy, TT2 4 weeks later
Additional boosters for full protection	TT3, TT4 and T5	Annually after previous dose
3–4 DTP doses in childhood	TT1 + TT2	During first pregnancy

5 doses provide long-lasting immunity

Immunization of pregnant women:

- **In those areas, any woman of childbearing age visiting a health facility should be screened and offered immunization, no matter what the reason for the visit.**

- **Non immunized women should receive at least 2 doses of tetanus toxoid according to the following schedule: the first at initial contact or as early as possible during pregnancy, the second 4 weeks after the first and preferably at least 2 weeks before delivery. A third dose could be given 6-12 months after the second, or during her next pregnancy.**

- **Additional 2 doses should be given at annual intervals when the mother is in contact with the health service or during her subsequent pregnancies.**

- **A total of 5 doses should protect the previously unimmunized woman through her entire childbearing period.**

- **Women whose infants have a risk of neonatal tetanus, but who themselves have received 3 or 4 doses of DTP as children, need only receive two doses of tetanus toxoid during their first pregnancy.**

SUMMARY

- **Tetanus is preventable; gaps in immunization persist**

Immunity:

a. Active: tetanus toxoid (≥10 years protection)

b. Passive: TIG

- **Disease does not confer immunity**

Prevention:

Health education, immunization, wound hygiene, sanitation

Specific Prevention

- Active vaccines: DTaP, DT, Td, TT
- Passive: TIG

Post-Exposure Prophylaxis

- Wound cleansing, metronidazole, TT booster, TIG when indicated

Neonatal Prevention

- Maternal immunization, skilled birth, aseptic cord care

TETANUS

A Deadly but Preventable Disease

Caused by
Clostridium tetani
Prevent it with
Vaccination!

• Symptoms

- Lockjaw (Jaw stiffness)
- Muscle spasms & stiffness
- Difficulty swallowing
- Fever & sweating

• Prevention

- Tetanus vaccine (every 10 years)
- Clean all wounds immediately
- Use sterile instruments
- Vaccinate pregnant mothers

Not spread person-to-person

Life-threatening if untreated!

Seek urgent care for deep/dirty wounds

Tetanus Toxoid Vaccination for Pregnant Women

Woman's Vaccination Status	Doses Needed	Timing of Doses
No previous tetanus vaccination	TT1 + TT2 (minimum)	TT1 during pregnancy, TT2 4 weeks later
Additional boosters for full protection	TT3, TT4 and T5	Annually after previous dose
3–4 DTP doses in childhood	TT1 + TT2	During first pregnancy

5 doses provide long-lasting immunity

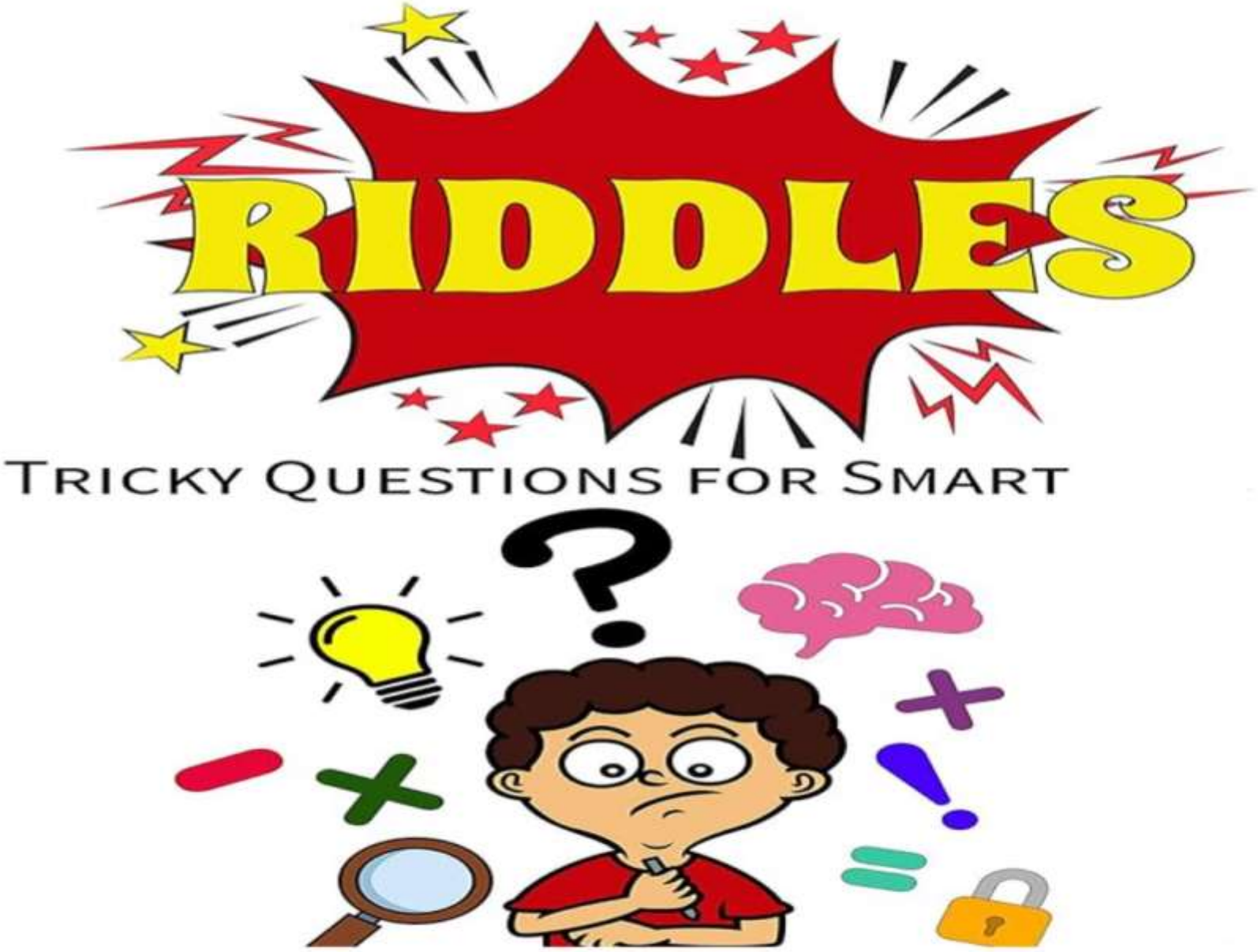
. 1What organism causes tetanus?

. 2Why doesn't tetanus confer immunity?

. 3Most common form of tetanus?

. 4Steps in PEP?

. 5Prevention of neonatal tetanus?


RIDDLES
TRICKY QUESTIONS FOR SMART

RIDDLE 1

I am a disease that does not spread from person to person.

Even if a patient is hospitalized, isolation is not required.

However, vaccination is still essential for the whole community.

Why is vaccination important if the disease is not contagious?

RIDDLE 2

A farmer suffers a deep puncture wound while working in the field.

He received his last tetanus vaccine 12 years ago.

The wound is small but contaminated with soil.

What is the most appropriate preventive action?

RIDDLE 3

Two patients arrive at the emergency department with wounds.

Patient A: fully immunized, last booster 3 years ago,
minor clean wound.

Patient B: unknown vaccination history, crush injury contaminated with soil.

Which patient requires passive immunization with TIG?

RIDDLE 4

A newborn develops muscle rigidity and difficulty feeding within the first week of life.

The delivery occurred at home, and the umbilical cord was cut with a non-sterile instrument.

The mother never received tetanus vaccination.

RIDDLE 5

A patient recovered from tetanus after intensive treatment.

Six months later, the physician recommeends tetanus vaccination.

Why is vaccination needed despite recovery from the disease?

RIDDLE 6

In a community with excellent vaccination coverage among children, tetanus cases still occasionally occur in elderly farmers.

What epidemiological explanation best accounts for this pattern?

RIDDLE 1

I am a disease that does not spread from person to person.

Even if a patient is hospitalized, isolation is not required.

However, vaccination is still essential for the whole community.

Why is vaccination important if the disease is not contagious?

Answer:

Because tetanus spores are widely present in the environment (soil, dust, animal feces), and everyone is at risk from contaminated wounds.

RIDDLE 2

A farmer suffers a deep puncture wound while working in the field.

He received his last tetanus vaccine 12 years ago.

The wound is small but contaminated with soil.

What is the most appropriate preventive action?

Answer:

Booster dose of tetanus toxoid because more than **10 years** have passed since the last vaccination.

RIDDLE 3

Two patients arrive at the emergency department with wounds.

Patient A: fully immunized, last booster 3 years ago, minor clean wound.

Patient B: unknown vaccination history, crush injury contaminated with soil.

Which patient requires passive immunization with TIG?

Answer:

Patient B, because of unknown immunization status and contaminated major wound.

RIDDLE 4

A newborn develops muscle rigidity and difficulty feeding within the first week of life.

The delivery occurred at home, and the umbilical cord was cut with a non-sterile instrument.

The mother never received tetanus vaccination.

Answer:

Patient B, tecause prevention is tetanus toxoid immunization of pregnant women and clean delivery practices.

RIDDLE 5

A patient recovered from tetanus after intensive treatment.

Six mexiths later, the physician recommends tetanus vaccination.

Why is vaccination needed despite recovery from the disease?

Answer:

Because natural infection does not produce protective immunity due to the extremely small amount of toxin required to cause disease.

RIDDLE 6

In community with reometal tetanus through a national public health program.

Which single intervention likely had the greatest impact?

Answer:

Immunization of women of childbearing ago with tetanus toxoid.

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